

Detailed derivations for human review

Companion to the finite-response RH formulation

Review record not yet completed

13 September 2026

Abstract

This file expands the chain of arguments needed for the condensed paper. It distinguishes the classical Weil criterion, unconditional operator calculations, and the unproved arithmetic matrix conjecture. It also gives finite-input error bounds suitable for rigorous computation. The source investigation is repository revision [efb908a](#); this companion is a new exposition, not an independently refereed validation of that manuscript. The human review should cover the hypotheses, domains, constants, quantifiers and citations as well as the displayed algebra. No verification of RH is claimed.

Contents

1	What must be reviewed	2
2	The arithmetic normalization, including the contact	2
3	Gamma expansion, source domain and cores	3
4	Neumann–Dirichlet comparison and the cutoff	4
5	Deriving and controlling the exact boundary correction	5
6	Matrix entries and finite-input tails	6
7	The finite response and both directions of the equivalence	8
8	The positive matrix comparison and the source interpretation	8
9	Residual identity, inverse order and Galerkin convergence	9
10	A fully specified finite-trial error allowance	10
11	The tolerance conjecture and its quantifiers	11
12	The exact prime discrepancy and a negative control	11
13	The length-one high-mode example	12
14	Human review record	13

1 What must be reviewed

The proposed short paper does not use the interacting vacuum metrics, physical residue normalization, the conditional tt^* equation, the remote delay construction, or the graph realization as premises. Those results remain in the longer investigation. The short paper needs only the following chain:

$$\begin{array}{c}
\text{Weil criterion and its exact arithmetic normalization} \\
\Downarrow \\
\text{closed logarithmic form and explicit boundary correction} \\
\Downarrow \\
\text{unconditional high-mode positivity and well-defined mixed block} \\
\Downarrow \\
\text{Schur identity on the full form domain} \\
\Downarrow \\
\text{RH} \iff S_n \succeq 0 \text{ for every } n \geq 1.
\end{array}$$

The remainder estimates provide a constructive way to approximate S_n ; they do not prove its sign. The positive shifted reference supplies an equivalent contraction statement but is not needed for the Schur identity.

Throughout, $X_L = L^2(0, L)$ over $\mathbb{C}$, inner products are antilinear first, $F = E_L f$ is the zero extension, and $\hat{F}(\tau) = \int F(x) e^{-i\tau x} dx$. The full-line shift is $U_d F(x) = F(x - d)$. Set $a_k = 2k + 1/2$ and write b for the scalar gamma multiplier, reserving B for the mixed operator. These differ from some letter choices in the long manuscript.

2 The arithmetic normalization, including the contact

For $g = F * \tilde{F}$, $\tilde{F}(x) = \overline{F(-x)}$, one has

$$g(0) = \|F\|^2, \quad g(-r) = \overline{g(r)}, \quad \text{Re } g(r) = \text{Re} \langle F, U_r F \rangle =: h_f(r).$$

With $n_\gamma(r) = e^{-r/2}/(1 - e^{-2r})$ for $r > 0$, the additive Weil functional, in the normalization of [1, Section 1.1], is

$$\begin{aligned}
\mathcal{W}(g) &= \int_{\mathbb{R}} 2 \cosh(x/2) g(x) dx - \sum_{p,m} (\log p) p^{-m/2} \{g(m \log p) + g(-m \log p)\} \\
&\quad - (\log(4\pi) + \gamma_E) g(0) - \int_0^\infty [g(r) + g(-r) - 2e^{-r/2} g(0)] n_\gamma(r) dr.
\end{aligned} \tag{2.1}$$

This equality is initially used on smooth compact support. The published criterion states that RH is equivalent to $\mathcal{W}(F * \tilde{F}) \geq 0$ for every complex $F \in C_c^\infty(\mathbb{R})$. This is the one external number-theoretic equivalence on which the short paper rests.

The gamma energy is

$$\begin{aligned}
K[F] &= \frac{1}{2} \int_{\mathbb{R}} \int_{\mathbb{R}} |F(x) - F(y)|^2 n_\gamma(|x - y|) dx dy \\
&= \int_0^\infty [2g(0) - g(r) - g(-r)] n_\gamma(r) dr.
\end{aligned} \tag{2.2}$$

Consequently the last line of (2.1) equals $K[F] + w_0\|F\|^2$, where

$$w_0 = -\log(4\pi) - \gamma_E - 2 \int_0^\infty (1 - e^{-r/2}) n_\gamma(r) dr.$$

The integral is convergent at zero and infinity. Expanding the denominator and integrating positive terms gives

$$\begin{aligned} \int_0^\infty (1 - e^{-r/2}) n_\gamma(r) dr &= \sum_{k \geq 0} \left(\frac{1}{2k + 1/2} - \frac{1}{2k + 1} \right) \\ &= \frac{1}{2} [\psi(1/2) - \psi(1/4)]. \end{aligned}$$

Using $\psi(1/2) = -\gamma_E - 2 \log 2$ proves

$$w_0 = \psi(1/4) - \log \pi = -\gamma_E - \pi/2 - 3 \log 2 - \log \pi. \quad (2.3)$$

This checks the negative contact without assigning it by a normalization of a positive state.

For the poles, convolution gives the kernel $2 \cosh((x - y)/2)$. On $(0, L)$, with $c(x) = \cosh((x - L/2)/2)$ and $s(x) = \sinh((x - L/2)/2)$, the identity $\cosh(u - v) = \cosh u \cosh v - \sinh u \sinh v$ gives $2cc^* - 2ss^*$. In particular the pole term has one positive and one negative direction, despite its pointwise positive kernel. The prime correlations vanish for $m \log p \geq L$. Thus the precise form is

$$Q_L[f] = K[E_L f] + w_0\|f\|^2 + 2|\langle c, f \rangle|^2 - 2|\langle s, f \rangle|^2 - 2 \sum_{m \log p < L} (\log p) p^{-m/2} \operatorname{Re} \langle F, U_{m \log p} F \rangle. \quad (2.4)$$

3 Gamma expansion, source domain and cores

The digamma partial fractions [3] give

$$\psi(z + it) - \psi(z) = \sum_{k \geq 0} \left[\frac{1}{k + z} - \frac{1}{k + z + it} \right].$$

At $z = 1/4$, taking real parts and substituting $t = \sqrt{s}/2$ yields

$$b(s) = \sum_{k \geq 0} \frac{2}{a_k} \frac{s}{a_k^2 + s}. \quad (3.1)$$

The series converges locally uniformly in $s \geq 0$. Its summands are nonnegative and increasing in s , and at least one is strictly increasing. Digamma asymptotics give $b(\tau^2) = \log |\tau| - \log 2 - \psi(1/4) + O(\tau^{-2})$. Thus $1 + b(\tau^2)$ is comparable above and below to $1 + \log(2 + |\tau|)$.

For $a > 0$, minimizing

$$\mathcal{E}_a(F, u) = \frac{2}{a} (\|F - u\|^2 + a^{-2} \|u'\|^2)$$

gives $\hat{u} = a^2 \hat{F} / (a^2 + \tau^2)$ and minimum $\int (2/a) \tau^2 / (a^2 + \tau^2) |\hat{F}|^2 d\tau / (2\pi)$. This can alternatively be represented by the output column

$$\left(\sqrt{2/a} H_{\mathbb{R}} (a^2 + H_{\mathbb{R}})^{-1} F, \sqrt{2a} \partial_x (a^2 + H_{\mathbb{R}})^{-1} F \right), \quad H_{\mathbb{R}} = -\partial_x^2.$$

Squaring the two components gives exactly the same multiplier. The orthogonal sum of these columns is a closed source: convergence of input and output identifies every bounded component,

and the square-summability of the limiting output is exactly membership in its maximal domain. That domain after zero extension is

$$\mathcal{D}_L = \{f : \int_{\mathbb{R}} \log(2 + |\tau|) |\widehat{E_L f}(\tau)|^2 d\tau < \infty\}. \quad (3.2)$$

The resulting form t_L is closed. Every prime shift is a contraction after compression, and the two pole functions are in $L^2(0, L)$, so their finite sum R_L is a bounded self-adjoint operator. Bounded perturbation gives a closed semibounded form $t_L + R_L$ and the operator $W_L = T_L + R_L$ on $\text{Dom } T_L$.

3.1 Smooth compact support is a form core

Translate to $(-L/2, L/2)$ for this proof. Given the zero extension F , let $D_r F(x) = r^{-1/2} F(x/r)$, $r < 1$. The Fourier transform is $r^{1/2} \widehat{F}(r\tau)$. Changing variable $\sigma = r\tau$ shows that the weighted norm uses $1 + \log(2 + |\sigma|/r)$. For r in, say, $[1/2, 1]$, this is uniformly comparable to the original weight. Dilations converge strongly as $r \uparrow 1$: first prove it on Schwartz functions by dominated convergence, then use their density in the weighted Fourier space and uniform boundedness. The support of $D_r F$ lies strictly inside the interval. Convolution with a smooth approximate identity of radius smaller than the support gap stays inside the interval, and its bounded Fourier multiplier converges to one in the same weighted norm. This proves the required core assertion. No pointwise trace at the interval endpoints is imposed on $\mathcal{D}_L$.

3.2 Compactness and the scalar truncation error

For a bounded set in the form norm, its Fourier mass outside $[-R, R]$ is bounded by a constant divided by $\log(2 + R)$. The map from a fixed Fourier band to $L^2(0, L)$ has a square-integrable kernel and is compact. Approximating the embedding by these compact maps shows that $\mathcal{D}_L \hookrightarrow X_L$ is compact. The associated operators therefore have compact resolvent. They can have finitely many negative eigenvalues; this is not their absence.

For numerical enclosure of $b(s)$, positivity and an integral test give

$$0 \leq b(s) - \sum_{k < J} \frac{2}{a_k} \frac{s}{a_k^2 + s} \leq s \left(\frac{2}{a_J^3} + \frac{1}{2a_J^2} \right). \quad (3.3)$$

Indeed $a_k = a_J + 2(k - J)$, and $\int_J^\infty 2(2x + 1/2)^{-3} dx = 1/(2a_J^2)$.

4 Neumann–Dirichlet comparison and the cutoff

For one mass, restricting a whole-line competitor u to $(0, L)$ drops the nonnegative exterior energy. Minimizing over all $H^1(0, L)$ competitors yields the Neumann resolvent. In the other direction, a Dirichlet competitor extends by zero to an admissible whole-line competitor. Thus, for every finite tower,

$$b_J(H_N) \leq T_{L,J} \leq b_J(H_D).$$

Monotone limits of the nonnegative forms preserve this order and its domain inclusions. In particular

$$b(H_N) \leq T_L \leq b(H_D). \quad (4.1)$$

Min–max, indexing eigenvalues from one, gives

$$b((j-1)\pi/L)^2 \leq \lambda_j(T_L) \leq b((j\pi/L)^2).$$

The logarithmic asymptotic then yields $\lambda_j(T_L) = \log(\pi j/(2L)) - \psi(1/4) + O_L(j^{-1})$.

For one prime p , put $d = \log p$ and $q = p^{-1/2}$. Express $x = t + kd$ with $0 < t < d$ and retain those integers k for which $0 < x < L$. This decomposes X_L into a direct integral of finite chain spaces. Their lengths are at most $h = \lceil L/d \rceil$. The complete off-diagonal correlation on a chain is the matrix with entries $q^{|i-j|}$ for $i \neq j$, zero on the diagonal. Its operator norm is bounded by the maximum absolute row sum $r_h(q)$. Therefore the negative prime contribution is bounded below by $-dr_h(q)I$. The assertion also holds when L/d is an integer; endpoint chains of exceptional length occur only on null sets.

Direct integration gives

$$\|c\|^2 = L/2 + \sinh(L/2), \quad \|s\|^2 = -L/2 + \sinh(L/2), \quad \langle c, s \rangle = 0.$$

Dropping $2cc^*$ and bounding $-2ss^*$ proves

$$W_L \geq b(H_N) - \beta_L I, \quad \beta_L = -w_0 + 2 \sinh(L/2) - L + \sum_{\log p < L} (\log p) r_{\lceil L/\log p \rceil}(p^{-1/2}). \quad (4.2)$$

An optional coarser prime term is $2 \sum_{m \log p < L} (\log p) p^{-m/2}$.

For $\delta_0 = 1/16$, define $N(L)$ as in the short paper and put $P = P_{N(L)}$, $Q = I - P$. The high form on $Q\mathcal{D}_L$ has the lower bound

$$H \geq \Lambda, \quad \Lambda e_j = d_j e_j, \quad d_j = b((j\pi/L)^2) - \beta_L \geq \delta_0 \quad (j \geq N).$$

Its inverse is legitimate before RH is assumed. The comparison uses the gamma operator and a crude bound on the signed remainder; it does not select low modes by assuming the spectrum of W_L is positive.

5 Deriving and controlling the exact boundary correction

For a mass $a > 0$, let $R_{f,a}$ be the compressed free resolvent and $R_{N,a}$ the Neumann resolvent. Their kernels are

$$R_{f,a}(x, y) = \frac{e^{-a|x-y|}}{2a}, \quad R_{N,a}(x, y) = \frac{\cosh(a \min(x, y)) \cosh(a(L - \max(x, y)))}{a \sinh(aL)}.$$

Both solve $(-\partial_x^2 + a^2)R = \delta_y$ in the open interval; the derivative jump is -1 . The second kernel has zero normal derivative at both endpoints. Expand its hyperbolic factors separately on $x \leq y$ and $x \geq y$. Subtraction gives

$$2a(R_{N,a} - R_{f,a}) = K_a = \frac{u_a u_a^* + v_a v_a^* + r_a(u_a v_a^* + v_a u_a^*)}{1 - r_a^2}, \quad (5.1)$$

where $u_a = e^{-ax}$, $v_a = e^{-a(L-x)}$, $r_a = e^{-aL}$. The coefficient matrix is $(1 - r_a^2)^{-1} \begin{pmatrix} 1 & r_a \\ r_a & 1 \end{pmatrix}$, with positive eigenvalues $1/(1 - r_a)$ and $1/(1 + r_a)$. The energy summand is $(2/a)I - 2aR$, so the free compressed energy is the Neumann energy plus K_a , with the positive sign.

5.1 Uniform boundedness of the infinite sum

Let $n(t) = \sum_{k \geq 0} e^{-akt} = e^{-t/2}/(1 - e^{-2t})$. In the kernel of K_a , the first $u_a u_a^*$ and $v_a v_a^*$ image terms sum to $n(x+y) + n(2L - x - y)$. Every additional diagonal image has an extra factor e^{-2aL} ; the mixed images have path length at least L . Thus the remainder is bounded, for example by

$$\frac{2[n(2L) + n(L)]}{1 - e^{-L}}.$$

As $n(t) - (2t)^{-1}$ is bounded on $(0, 2L]$, the sum is

$$K_L = \frac{1}{2}(C_0 + C_L) + \text{a bounded-kernel operator}, \quad (5.2)$$

where $C_0(x, y) = (x + y)^{-1}$ and $C_L(x, y) = (2L - x - y)^{-1}$. Using weight $x^{-1/2}$, the identity

$$\int_0^\infty \frac{y^{-1/2}}{x + y} dy = \pi x^{-1/2}$$

proves $\|C_0\| \leq \pi$ by the weighted Schur test; reflection treats C_L . The same kernel majorant bounds every positive partial sum. The monotone bounded sequence $K^{(J)} = \sum_{k < J} K_{a_k}$ converges strongly to a bounded positive K_L .

For any $f \in X_L$, finite-tower energies satisfy $t_{L,J}[f] = \langle f, b_J(H_N)f \rangle + \langle f, K^{(J)}f \rangle$. The last term has a bounded limit for every f . Hence the two increasing energy limits are finite for exactly the same f , and their closed forms differ by K_L . This proves

$$T_L = b(H_N) + K_L, \quad \text{Dom } T_L = \text{Dom } b(H_N), \quad \mathcal{D}_L = \text{Dom } b(H_N)^{1/2}. \quad (5.3)$$

Spectral partial sums of $b(H_N)$ are an operator core; bounded perturbation preserves that fact. In particular all finite cosine sums belong to $\text{Dom } W_L$, not merely to its form domain.

5.2 Why the boundary operator is not compact

Split $(0, L)$ at $L/2$. In (5.2), all off-diagonal blocks and the opposite endpoint kernel within each diagonal block are bounded kernels. They are Hilbert–Schmidt. Modulo compact operators, K_L is the direct sum of two half-Carleman operators. Thus its essential norm is at most $\pi/2$.

For the reverse bound take $\phi_R(t) = t^{-1/2}1_{[1,R]}(t)/\sqrt{\log R}$ on $(0, \infty)$. Changing to $t = e^u$ transforms the Carleman kernel to convolution by $k(u - v) = [2 \cosh((u - v)/2)]^{-1}$. Hence its Rayleigh quotient is

$$\int_{-\log R}^{\log R} \left(1 - \frac{|v|}{\log R}\right) k(v) dv \longrightarrow \int_{\mathbb{R}} k(v) dv = \pi.$$

For fixed R , dilate into $(\epsilon, \epsilon R) \subset (0, L/2)$. The unit vectors tend weakly to zero and keep their Carleman quotient, while every compact quadratic term tends to zero. First let $\epsilon \downarrow 0$, then $R \rightarrow \infty$. This gives the reverse essential-norm bound and proves

$$\|K_L\|_{\text{ess}} = \pi/2, \quad \|K_L - K^{(J)}\| \geq \pi/2 \quad \text{for every finite } J.$$

This does not conflict with compact resolvent of the unbounded T_L . The needed Carleman facts are proved here; [5] gives broader context.

6 Matrix entries and finite-input tails

Let $\omega_j = j\pi/L$ and $e_j = \nu_j \cos(\omega_j x)$. Integration gives

$$\langle u_a, e_j \rangle = \frac{\nu_j a (1 - (-1)^j e^{-aL})}{a^2 + \omega_j^2}, \quad \langle v_a, e_j \rangle = (-1)^j \langle u_a, e_j \rangle.$$

Inserting this into (5.1) cancels the denominator and yields

$$K_a e_j = \frac{\nu_j a}{a^2 + \omega_j^2} (u_a + (-1)^j v_a). \quad (6.1)$$

Taking a second inner product gives zero for opposite parities, and for equal parities gives

$$\langle e_i, K_L e_j \rangle = 2\nu_i \nu_j \sum_{k \geq 0} \frac{a_k^2 (1 - (-1)^j e^{-a_k L})}{(a_k^2 + \omega_i^2)(a_k^2 + \omega_j^2)}. \quad (6.2)$$

The parity separation follows also from reflection about $L/2$. It is useful for calculation, but the short paper does not rely on an odd-only variant of the Weil criterion.

Since $\|u_a\|, \|v_a\| \leq (2a)^{-1/2}$, equation (6.1) gives $\|K_a e_j\| \leq \sqrt{2} \nu_j a^{-3/2}$. The integral test yields $\sum_{k \geq J} a_k^{-3/2} \leq a_J^{-3/2} + a_J^{-1/2}$. The operator norm on the first h columns is at most their Hilbert–Schmidt norm. Using $\sum_{j < h} \nu_j^2 = (2h - 1)/L$ proves

$$\|(K_L - K^{(J)})P_h\| \leq \sqrt{\frac{2(2h - 1)}{L}} (a_J^{-1/2} + a_J^{-3/2}) =: \eta_{h,J}. \quad (6.3)$$

Only the input dimension h is fixed in this limit. It does not imply operator-norm convergence on the full space.

6.1 Prime and pole entries without an unspecified kernel

Define

$$\mathcal{I}(\lambda, \phi; a, b) = \begin{cases} [\sin(\lambda b + \phi) - \sin(\lambda a + \phi)]/\lambda, & \lambda \neq 0, \\ (b - a) \cos \phi, & \lambda = 0. \end{cases}$$

For $0 < d < L$, product-to-sum gives

$$\begin{aligned} t_{ij}(d) := \langle e_i, V_d e_j \rangle &= \frac{\nu_i \nu_j}{2} [\mathcal{I}(\omega_i - \omega_j, \omega_j d; d, L) \\ &\quad + \mathcal{I}(\omega_i + \omega_j, -\omega_j d; d, L)]. \end{aligned} \quad (6.4)$$

The adjoint entry is $t_{ji}(d)$. This handles the zero-frequency cases by the second branch of $\mathcal{I}$.

For $\sigma \neq 0$ set

$$J_j(\sigma) = \int_0^L e^{\sigma x} \cos(\omega_j x) dx = \frac{\sigma [(-1)^j e^{\sigma L} - 1]}{\sigma^2 + \omega_j^2}.$$

Then

$$\begin{aligned} c_j &:= \langle e_j, c \rangle = \frac{\nu_j}{2} [e^{-L/4} J_j(1/2) + e^{L/4} J_j(-1/2)], \\ s_j &:= \langle e_j, s \rangle = \frac{\nu_j}{2} [e^{-L/4} J_j(1/2) - e^{L/4} J_j(-1/2)]. \end{aligned}$$

Thus the entire cosine matrix is

$$w_{ij} = (b_j + w_0) \delta_{ij} + k_{ij} + 2c_i c_j - 2s_i s_j - \sum_{m \log p < L} (\log p) p^{-m/2} [t_{ij}(m \log p) + t_{ji}(m \log p)]. \quad (6.5)$$

All entries are real, but the quadratic-form assertions apply to complex inputs. For residual norms, the full columns are also explicit:

$$\begin{aligned} W_L e_j &= (b_j + w_0) e_j + K_L e_j + 2c_j c - 2s_j s \\ &\quad - \sum_{d=m \log p < L} (\log p) p^{-m/2} [1_{x>d} e_j(x - d) + 1_{x<L-d} e_j(x + d)]. \end{aligned} \quad (6.6)$$

Finite mass sums in (6.1) turn these into finite sums of exponentials, trigonometric functions, and their interval restrictions. Products and squared norms are elementary integrals on the finitely many overlap intervals. Equations (3.3) and (6.3) supply the missing infinite tails.

7 The finite response and both directions of the equivalence

Because $PX_L \subset \text{Dom } W_L$, both $A = PW_L P$ and $B = QW_L P$ are bounded on the finite low space. Restriction of the closed form defines $H \geq \delta_0 I$ on QX_L . Its operator domain is $Q \text{Dom } b(H_N)$, by (5.3). For $p = Pf$ and $h = Qf$ the form is

$$Q_L[p + h] = \|H^{1/2}h\|^2 + 2\text{Re}\langle h, Bp \rangle + \langle p, Ap \rangle.$$

As $H^{-1}Bp \in \text{Dom } H$, expand the square to obtain

$$Q_L[p + h] = \|H^{1/2}(h + H^{-1}Bp)\|^2 + \langle p, S_L p \rangle, \quad S_L = A - B^*H^{-1}B. \quad (7.1)$$

The expansion uses $H^{1/2}H^{-1} = H^{-1/2}$ and the closed form pairing. It does not require every h to be in $\text{Dom } H$.

If $S_L \succeq 0$, equation (7.1) gives $Q_L \geq 0$ on its full form domain. Conversely choose $h = -H^{-1}Bp$. The high square vanishes and $Q_L[p + h] = \langle p, S_L p \rangle$, so positivity of Q_L forces positivity of S_L . A negative vector of S_L therefore produces a negative form-domain input. Approximation by the smooth core produces a smooth negative input.

Under RH, the classical Weil criterion gives positivity on the smooth core; continuity in the shifted form norm extends it to $\mathcal{D}_L$. Thus $S_n \succeq 0$ for every integer $n \geq 1$. Conversely, every compactly supported test function can be translated into $(0, n)$ for an integer n . Translation changes its Fourier transform by a phase and preserves every translated correlation. The pole kernel $2 \cosh((x - y)/2)$ also depends only on the difference, so the entire form is invariant. Zero extension to a larger interval preserves the same form and merely adds inactive translations with zero overlap. Positivity of every S_n now implies the global Weil criterion, hence RH.

This argument uses only a cofinal sequence of supports. It requires neither source compatibility nor a uniform full-form gap. It also shows that an arbitrary finite list of successful lengths cannot finish the proof.

8 The positive matrix comparison and the source interpretation

For $\alpha = \beta_L + 1$, the operator $T = W_L + \alpha I \geq I$ is positive independently of RH. Its block matrix is

$$T = \begin{pmatrix} A + \alpha I & B^* \\ B & H + \alpha I \end{pmatrix}.$$

Let $V = (H + \alpha I)^{-1}$. Completing the shifted high square yields

$$G = A + \alpha I - B^*VB \geq I.$$

To check the bound, minimize $\langle p + h, T(p + h) \rangle \geq \|p\|^2 + \|h\|^2$ over h at fixed p . Define

$$D = \alpha I + B^*(H^{-1} - V)B = \alpha[I + B^*VH^{-1}B] \geq \alpha I.$$

The product VH^{-1} is positive because both factors are functions of the same H ; arbitrary products of positive operators would not justify this step. Direct subtraction gives $G - D = S_L$. Congruence by $G^{-1/2}$ therefore proves

$$S_L \succeq 0 \iff G^{-1/2}DG^{-1/2} \preceq I. \quad (8.1)$$

Both G and D can change with the permissible positive shift; their difference cannot. This rules out tuning a source normalization to improve the arithmetic sign.

Abstractly, any positive source $\mathcal{A}$ for T has a low output orthogonal to its high range, with Gram G . The defect output has Gram D . A contraction taking the former output to the latter exists precisely when $D \preceq G$. Existence alone is therefore the same conjecture. A microscopic physical law constructing such a contraction from specified interactions would be additional structure, whose necessity under RH is not established by this equivalence.

For a simple control, take $W = \begin{pmatrix} 1/2 & 1 \\ 1 & 1 \end{pmatrix}$ and $\alpha = 1$. Then $H = 1$, $V = 1/2$, $G = 1$, $D = 3/2$, and $S = -1/2$. The positive shifted reference and the positive high block coexist with negative full-form energy. The conjectural ordering is an essential step.

9 Residual identity, inverse order and Galerkin convergence

For a trial map $Y : PX_L \rightarrow \text{Dom } H$, let $R = B - HY$. Expand without reordering any noncommuting factors:

$$R^* H^{-1} R = B^* H^{-1} B - B^* Y - Y^* B + Y^* H Y.$$

This proves

$$B^* H^{-1} B = B^* Y + Y^* B - Y^* H Y + R^* H^{-1} R. \quad (9.1)$$

The form inequality $H \geq \Lambda > 0$ reverses for inverses: $H^{-1} \leq \Lambda^{-1}$. One proof uses

$$\langle z, H^{-1} z \rangle = \sup_v \{2 \operatorname{Re} \langle z, v \rangle - \|H^{1/2} v\|^2\} \leq \sup_v \{2 \operatorname{Re} \langle z, v \rangle - \|\Lambda^{1/2} v\|^2\}.$$

The form-domain inclusion makes the supremum comparison valid. Thus $R^* H^{-1} R \leq R^* \Lambda^{-1} R$.

Let $E_K = P_{N,K}$ and solve $E_K H E_K Y_K = E_K B$ in $E_K X_L$. The finite coefficient matrix is at least $\delta_0 I$ and hence invertible. Multiplying by Y_K^* gives

$$Y_K^* H Y_K = Y_K^* B = B^* Y_K, \quad E_K (B - H Y_K) = 0.$$

The residual R_K has no modes below K , so

$$A - B^* Y_K - d_K^{-1} R_K^* R_K \preceq S_L \preceq A - B^* Y_K. \quad (9.2)$$

The lower expression, denoted $L_{L,K}$, includes the entire residual Gram. Omitting it leaves an upper bound.

For convergence, each exact solution column $u = H^{-1} B e_j$ minimizes the coercive energy with that source. The Galerkin solution u_K is its orthogonal projection in the H -form inner product onto the trial space. The cosine core gives $\|H^{1/2}(u - u_K)\| \rightarrow 0$; in particular $u_K \rightarrow u$ in L^2 . There are finitely many source columns, so this is operator-norm convergence on the input space. Write $H = \Lambda + E$ with E bounded, using (5.3). Then

$$R_K = (I - P_K)(B - E Y_K).$$

Here $(I - P_K) \Lambda Y_K = 0$ and the other residual rows vanish by Galerkin orthogonality. Since $E Y_K \rightarrow E H^{-1} B$ in norm and the input space is finite dimensional, $R_K \rightarrow 0$ in norm. Consequently the gap between the two matrices in (9.2) tends to zero.

For ordered inverse approximations the direction of the bound is easy to reverse accidentally. If $F_\ell \preceq H^{-1}$, then $A - B^* F_\ell B \succeq S_L$. For example with $H = 1$, $\alpha = 1$, $V = 1/2$, $F_\ell = \sum_{j=0}^\ell 2^{-j-1} = 1 - 2^{-\ell-1}$ and $A = 1 - 2^{-\ell-2}$, one has

$$A - F_\ell = 2^{-\ell-2} > 0, \quad A - H^{-1} = -2^{-\ell-2} < 0.$$

Positive approximate upper matrices cannot certify the target.

10 A fully specified finite-trial error allowance

This section makes the word “approximable” operational. It is a consequence of the boundary column estimate and the residual identity, with an explicit allowance for a finite mass approximation. It is not an executed positivity certificate for any arithmetic matrix.

Choose $K > N$ and any matrix $Y : PX_L \rightarrow P_{N,K}X_L$, for example one with rational coefficients. No exact inverse or exact Galerkin solution is required. Set $V_0 = P - Y$ as a map from the low coefficient space to P_KX_L . Define

$$W^{(J)} = b(H_N) + K^{(J)} + R_L, \quad C_J = V_0^* W^{(J)} V_0, \quad \tilde{R} = QW^{(J)} V_0.$$

The true residual is $R = QW_L V_0 = B - HY$. Positivity of $K_L - K^{(J)}$ gives $V_0^* W_L V_0 \succeq C_J$, while (6.3) gives

$$\|\Lambda^{-1/2}(R - \tilde{R})\| \leq \frac{\eta_{K,J}\|V_0\|}{\sqrt{\delta_0}} =: \epsilon_{K,J,Y}, \quad \|V_0\|^2 = 1 + \|Y\|^2. \quad (10.1)$$

The last equality uses orthogonality of the low and trial high spaces.

For a chosen $M > N$, let $r_j = e_j^* \tilde{R}$ and define the finite matrix

$$T_M = \sum_{j=N}^{M-1} \frac{r_j^* r_j}{d_j} + \frac{1}{d_M} \left(\tilde{R}^* \tilde{R} - \sum_{j=N}^{M-1} r_j^* r_j \right). \quad (10.2)$$

It is positive and bounds $\tilde{R}^* \Lambda^{-1} \tilde{R}$ from above, by Parseval and monotonicity of d_j . All its quantities are finite integrals of the explicitly given functions in (6.6), with $K^{(J)}$ in place of K_L . In particular the full residual norm is kept even if only finitely many cosine coefficients are individually computed.

For operators X and Z with $\|Z\| \leq \epsilon$,

$$(X + Z)^*(X + Z) \preceq X^* X + (2\|X\|\epsilon + \epsilon^2)I.$$

Apply this with $X = \Lambda^{-1/2} \tilde{R}$ and use $\|X\| \leq \sqrt{\|T_M\|}$. Equations (9.1) and (10.1) then prove the rigorous lower bound

$$S_L \succeq C_J - T_M - \left(2\sqrt{\|T_M\|} \epsilon_{K,J,Y} + \epsilon_{K,J,Y}^2 \right) I. \quad (10.3)$$

This uses only finite matrices, integrals of finite sums, and explicit scalar error bounds. Extra numerical rounding or integration uncertainty must be subtracted as an additional scalar operator-norm allowance. If entrywise Hermitian errors are bounded by $e_{ij} \geq 0$, the maximal row sum of the symmetric error majorant bounds their operator norm. One may use rational upper bounds for all norms and lower bounds for the positive d_j and δ_0 .

The special-function values b_j can be enclosed by (3.3). Exponentials, logarithms and trigonometric values in the finite entries admit convergent elementary series with rational remainder estimates. At integer $L = n$, the active prime set is determined by $p^m < e^n$; there is no equality because e^n is transcendental. Finite enumeration is therefore well-defined, though potentially expensive.

For fixed trial Y and K , let $J \rightarrow \infty$ and refine scalar enclosures; then $\epsilon_{K,J,Y} \rightarrow 0$, $C_J \rightarrow V_0^* W_L V_0$, and the residual approximations converge in L^2 . Exact Galerkin trials with $K \rightarrow \infty$ have vanishing true residual by the preceding section. Approximating their finite coefficients by rationals and taking sufficiently many masses therefore makes (10.3) as close to S_L as needed. This proves availability of finite error control, not a bound on the cost or a sign for the resulting matrix.

11 The tolerance conjecture and its quantifiers

Define $L_{n,K}$ by (9.2). The alternate conjecture is

$$\forall n, m \in \mathbb{N} \quad \exists K > N(n) : \quad L_{n,K} + 2^{-m}I \succ 0. \quad (11.1)$$

If $S_n \succeq 0$, norm convergence gives a K such that $\|L_{n,K} - S_n\| < 2^{-m}$. Thus the displayed matrix is strictly positive. Conversely a witness K gives $S_n \succeq L_{n,K} \succ -2^{-m}I$. Since this holds for every m , each eigenvalue of S_n is nonnegative. Combining with Section 7 proves that (11.1) is equivalent to RH.

The order of quantifiers matters. Neither a single K for all m nor a uniform K or N for all n is asserted. Replacing “every m ” by one small tolerance would be weaker than positivity. Requiring an unshifted strict finite certificate for each n is stronger on its face and would need a separate strict-positivity argument to claim equivalence. The tolerance formulation avoids this unnecessary extra premise.

The strict margin in (11.1) also permits replacing the exact matrix by a sufficiently accurate certified lower enclosure such as (10.3). Under $S_n \succeq 0$, choose all approximation errors less than a fixed fraction of 2^{-m} . The existence argument then has slack. It remains necessary to prove the arithmetic inequalities for all n, m ; the generic convergence theorem is not that proof.

12 The exact prime discrepancy and a negative control

Let $h_f(r) = \text{Re}\langle F, U_r F \rangle$ and $E(r) = \psi_{\text{Ch}}(e^r) - e^r + 1$. The prime term is $-2 \int_0^L e^{-r/2} h_f(r) d\psi_{\text{Ch}}(e^r)$. The pole term is $4 \int_0^L \cosh(r/2) h_f(r) dr$. Subtracting the smooth prime density $e^r dr$ from this pole expression leaves $2 \int_0^L e^{-r/2} h_f(r) dr$, the form of the kernel $e^{-|x-y|/2}$. The $a_0 = 1/2$ gamma summand is $4I$ minus precisely that kernel, since $2a_0 R_{f,a_0}$ has kernel $e^{-|x-y|/2}$. Thus

$$Q_L[f] = t_{\geq 1, L}[f] + (4 + w_0)\|f\|^2 - 2 \int_0^L e^{-r/2} h_f(r) dE(r). \quad (12.1)$$

Autocorrelations of L^2 functions are continuous, and the finite-interval measure has finite variation. This is a bounded form contribution, so the identity extends from the smooth core to $\mathcal{D}_L$.

For smooth compact support, $E(0) = 0$ and $h_f(L) = 0$ remove both boundary terms in Stieltjes integration by parts. The last term becomes $2 \int_0^L E(r) e^{-r/2} (h'_f(r) - h_f(r)/2) dr$. No such derivative is assumed to be an ordinary function on the whole logarithmic domain.

If the prime measure is replaced by its leading density, then $E = 0$. For a zero extension in $H^1(\mathbb{R})$,

$$t_{\geq 1, L}[f] \leq \sum_{k \geq 1} \frac{2}{a_k^3} \|F'\|^2 \leq \frac{26}{125} \|F'\|^2.$$

The last fraction is $2/(5/2)^3 + 1/[2(5/2)^2]$ by an integral tail. Equation (2.3) implies $w_0 < -5$. To check the elementary inequality, use $\gamma_E > 1/2$, $\pi > 3$, $\log 2 > 2/3$ and $\log \pi > 1$. For example $\gamma_E > H_8 - \log 9 > H_8 - 11/5 = 29/56 > 1/2$; $e^{11/10} > 3$ follows from its first six positive series terms, and $e < 3$ gives $\log \pi > 1$.

On $(0, 2)$, $f(x) = \sin(\pi x/2)$ has norm one, vanishes at the endpoints, and $\|F'\|^2 = \pi^2/4$. Hence

$$Q_2^{\text{dens}}[f] < -1 + \frac{26}{125} \frac{(22/7)^2}{4} = -\frac{2979}{6125}.$$

Approximation in $H_0^1(0, 2)$ preserves negativity and gives smooth compact support. This shows why simply discarding the signed prime discrepancy does not provide a route to the matrix ordering. It does not rule out methods that use stronger, signed prime information.

13 The length-one high-mode example

At $L = 1$ only $p = 2, m = 1$ is active. The chain has two sites, so

$$\beta_1 = -w_0 + 2 \sinh(1/2) - 1 + \frac{\log 2}{\sqrt{2}}.$$

The longer manuscript supplies the rational bounds

$$\begin{aligned} \gamma_E &< 0.58, \quad \pi < 22/7, \quad \log 2 < 0.7, \\ \log \pi &< 23/20, \quad 2 \sinh(1/2) - 1 < 43/1000, \quad \log 2/\sqrt{2} < 1/2. \end{aligned}$$

They give

$$\beta_1 < \frac{58}{100} + \frac{11}{7} + \frac{21}{10} + \frac{23}{20} + \frac{43}{1000} + \frac{1}{2} = \frac{41611}{7000}.$$

These constants may be checked using rational series: the atanh series $\log 2 = 2 \sum_{j \geq 0} 3^{-(2j+1)}/(2j+1)$ gives its upper and lower bounds; $\gamma_E < H_{256} - \log 256$ gives the displayed Euler-constant bound with $\log 2 > 6931/10000$; positive exponential sums and geometric tail bounds give the remaining logarithm and sinh bounds. For $\pi < 22/7$ use $22/7 - \pi = \int_0^1 x^4(1-x)^4/(1+x^2)dx > 0$.

Since $\pi > 3$ and b is increasing,

$$b((4\pi)^2) > \sum_{k=0}^{63} \frac{2}{a_k} \frac{144}{a_k^2 + 144} > \frac{3007}{500} > \frac{41611}{7000} + \frac{1}{16}.$$

The two final comparisons use exact fractions. Thus $N = 4$ is an admissible cutoff. This proves $N(1) \leq 4$ for the canonical minimum; it is not necessary to prove minimality. When using this certificate, one may deliberately use the four-dimensional low space. No arithmetic Schur-matrix entries or sign certificate are inferred from these scalar inequalities.

14 Human review record

Mark an item complete only after checking its proof and any cited premise. This list starts unreviewed. A mathematical question should be recorded with the equation number and the needed correction, then resolved before that item is marked complete.

Required check	Reviewer / date
Weil normalization and classical equivalence, Section 2	_____
Closed logarithmic form and smooth core, Section 3	_____
Neumann comparison and all-prime cutoff, Section 4	_____
Green-kernel sign and equality of domains, Section 5	_____
Carleman essential norm and finite-input tails, Sections 5–6	_____
Prime/pole coefficients and all mixed terms, Section 6	_____
Schur identity and both RH implications, Section 7	_____
Positive reference and shift cancellation, Section 8	_____
Residual inequality and Galerkin convergence, Section 9	_____
Finite-trial error propagation, Section 10	_____
Tolerance quantifiers and null directions, Section 11	_____
Discrepancy sign and rational control, Sections 12–13	_____
Reference accuracy, prior work and claim of novelty	_____
Final text and disclosure match the review actually done	_____

Completing these items validates the proposed reduction only to the extent of the actual human checks; it does not prove the open conjecture and does not constitute independent peer review. The original package’s 59 algebra checks reproduce, but they do not certify the analytic arguments above. The physical portions omitted from the short paper need their own review if they are later submitted as separate mathematical claims.

Preparation note

OpenAI’s Codex assisted substantially with the analysis, calculations, code and writing of this companion. The finite tolerance formulation and the explicit finite-trial allowance are developed here from the residual and column estimates in the source investigation. They are supplied with proofs for review, not represented as externally checked results. No human review has yet been recorded in this file.

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- [6] *Arithmetic ground-state geometry: Physical caps, arithmetic sources and analytical sign tests*, working manuscript in the [shifted-zeta-positivity](#) project, September 2026, [repository revision efb908a](#), especially Sections 10, 14 and 15.